

Part I

SUMMARY of Applications of Integrals

Summary

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SUMMARY: Velocity, Speed, Position, Displacement, Distance

Position, $s(t)$, of an object at time t : $s(t) = s(a) + \int_a^t v(z)dz$

Displacement, $\Delta s = s(b) - s(a)$,
of an object over the time interval $[a, b]$: $s(b) - s(a) = \int_a^b v(t)dt$

Distance traveled by an object
over the time interval $[a, b]$: $\int_a^b |v(t)|dt$

SUMMARY: Rate of Accumulation, Amount, Change in the Amount

The amount, $A(t)$, of some
substance/population at the time t : $A(t) = A(a) + \int_a^t A'(z)dz$

The amount, $A(b)$,
over the time interval $[a, b]$: $A(b) = A(a) + \int_a^b A'(t)dt$

The change in the amount, $A(b) - A(a)$,
over the time interval $[a, b]$: $A(b) - A(a) = \int_a^b A'(t)dt$

SUMMARY: Average Value, the Mean Value Theorem for Integrals

Average value $\bar{f}$,
of the function f on the interval $[a, b]$: $\bar{f} = \frac{1}{b-a} \int_a^b f(x)dx$

Mean Value Theorem for Integrals

Let f be continuous on $[a, b]$. There exists a value c in (a, b) such that

Summary

$$\frac{1}{b-a} \int_a^b f(x) dx = f(c)$$

Note 1: $f(c) = \bar{f}$, the average value of f on $[a, b]$.

Note 2: The net area of the region between the curve $y = f(x)$ and the x -axis is given by

$$\int_a^b f(x) dx = f(c)(b-a)$$

Part II

Recitation Questions

Problem 1

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Problem 1 Solve the following word problems:

- (a) The velocity function for an object moving along a line east/west is given by $v(t) = -t^2 + 4t - 3$ feet per minute.
 - (i) Find the total displacement the object traveled from 2 minutes to 6 minutes (assume east is positive).
 - (ii) Find the total distance the object traveled from 2 minutes to 6 minutes.
 - (iii) Suppose that the object's position 2 minutes into the trip is 5 feet of a placement marker. What is its position (relative to the placement marker) at 6 minutes.
- (b) Sammy the Snail sets up camp in the median of I-70 and, starting at noon and ending at 6pm, hikes back and forth along the highway. He starts his hike at his campsite. His velocity at time t hours (after noon) is given by $v(t) = (t - 2)(t - 5)$ inches per hour. Find the total distance Sammy travelled on his hike.

Problem 2

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Problem 2

Problem 2 Suppose that $r(t) = r_0 e^{-kt}$ (with $k > 0$) is the rate at which a nation extracts oil. The current rate of extraction is $r(0) = 10^7$ barrels/yr. Also assume that the estimate of the total oil reserve (ie, the amount of oil remaining beneath the ground in this country) is 2×10^9 barrels.

- (a) Find $A(t)$, the total amount of oil extracted by the nation after t years.
- (b) Evaluate $\lim_{t \rightarrow \infty} A(t)$ and explain the meaning of this limit.
- (c) Find the minimum constant k for which the total oil reserves will last forever.
- (d) Suppose that the constant k is half the minimum value found in part (c). When will the nation deplete its oil reserve?

Problem 3

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Problem 3

Problem 3 Assume that the rate of change (in dollars per day) of the price of shares of stock in the WeSaySo Company (with t in days) is modeled by the equation $r(t) = -3t^2 + 30t - 63$ (note that this is technically a discrete function, but prices change so often with stocks that modeling this with a continuous function makes sense). Assume also that the price of a share of stock on day 1 (i.e., $t = 1$) is \$51. Answer the following questions:

- (a) Find the rate of change of price at $t = 5$.
- (b) Find the price of a share of stock at $t = 5$.
- (c) How fast is the rate of change of price changing at $t = 5$?
- (d) How much did the price of a share of stock change in the first 6 days (i.e., on $[1, 6]$)?
- (e) What was the greatest rate of change of price during the first 6 days (i.e., on $[1, 6]$)? [3]
- (f) What was the greatest price of a share of stock during the first 6 days (i.e., on $[1, 6]$)? [3]

Problem 4

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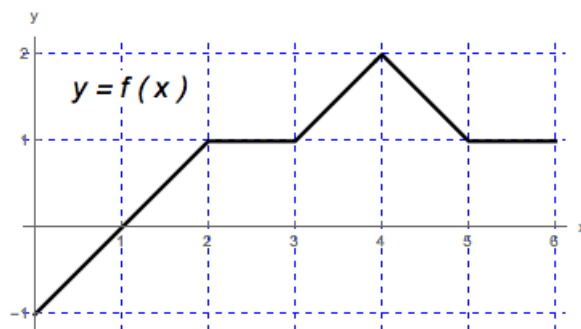
Problem 4

Problem 4 A cup of coffee has temperature $20 + 75e^{-.02t}$ degrees (Celsius) t minutes after being poured into a cup. What is the average temperature of the coffee during the first half hour?

Problem 5

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Problem 5 The graph of a function f defined on the interval $[0, 6]$ is given in the figure.



- Compute the net area of the region between the graph of f and the x -axis, on the interval $[0, 6]$. [3]
- Draw a rectangle with base on the x -axis, $0 \leq x \leq 6$, whose area is equal to the net area in part (a).
- In the figure, mark a point c in $(0, 6)$ such that $f(c)$ is the height of the rectangle from part (b).
- Using the figure and parts (a-c), what is the relationship between the rectangle from part (b), the net area from part (a), and the average value of f on $[0, 6]$? [3]

Problem 6

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Problem 6

Problem 6 Find all points at which the given function equals its average value on the given interval.

(a) $f(x) = e^x$ $[0, 4]$

(b) $g(x) = \frac{\pi}{4} \sin(x)$ $[0, \pi]$

Problem 7

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Problem 7

Problem 7 Find the average value of the function $g(t) = 4te^{-t^2}$ on the interval $[0, 3]$.